

Bridging the Network and Link Layers

From Control Plane Routing to Link-Layer Frames

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IT 213: Networking

Winter 2026

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Recap: The Network Layer and the Control Plane

In the previous chapters, we learned that the network layer provides a communication service between any two network hosts.

- **The Control Plane:** This is the network-wide logic. Routing protocols (like OSPF and BGP) act as the "brains," computing the end-to-end paths across the globe.
- **The Travel Agent Analogy:** Consider a tourist traveling from Princeton to Lausanne.
- The travel agent decides the tourist should take a limo to JFK, a plane to Geneva, and a train to Lausanne.
- In this analogy, the tourist is the datagram, and the travel agent is the routing protocol.

The Transition: Crossing the Physical Links (Section 6.1)

While the routing protocol plans the trip, it does not actually drive the vehicles.

- Between the two hosts, datagrams travel over a series of communication links, passing through a series of packet switches (switches and routers).
- Each of the transportation segments (the limo, the plane, the train) represents a single link.
- The basic service of any link layer is to move a datagram from one node to an adjacent node over a single communication link.
- How does this happen? The transmitting node encapsulates the datagram in a link-layer frame and transmits the frame into the link.

Core Link Layer Services (Section 6.1.1)

Different link-layer protocols (like Ethernet or Wi-Fi) offer different services, but they share common functions:

- **Framing:** Almost all link-layer protocols encapsulate each network-layer datagram within a link-layer frame before transmission over the link.
- A frame consists of a data field, in which the network-layer datagram is inserted, and a number of header fields.
- **Link Access:** A medium access control (MAC) protocol specifies the rules by which a frame is transmitted onto the link.

Where is the Link Layer Implemented? (Section 6.1.2)

Unlike the upper layers of the protocol stack, which are entirely software, the link layer spans the boundary between hardware and software.

- **The Network Adapter:** For the most part, the link layer is implemented on a chip called the network adapter, also sometimes known as a network interface controller (NIC).
- Much of a link-layer controller's functionality—like framing, link access, and error detection—is implemented in hardware.
- **The CPU:** Part of the link layer is implemented in software that runs on the host's CPU, responding to controller interrupts and passing datagrams up to the network layer.

MAC Addresses (Section 6.4.1)

To move frames across a local area network, we need a new type of address.

- In truth, it is not hosts and routers that have link-layer addresses but rather their adapters (that is, network interfaces) that have link-layer addresses.
- For most LANs, the MAC address is 6 bytes long.
- **Structure:** An adapter's MAC address has a flat structure and doesn't change no matter where the adapter goes.
- This is analogous to a person's social security number, whereas an IP address is hierarchical (like a postal address) and changes when the host moves.

The Address Resolution Protocol - ARP (Section 6.4.1)

Because there are both network-layer addresses (IP addresses) and link-layer addresses (MAC addresses), there is a need to translate between them.

- For the Internet, this is the job of the Address Resolution Protocol (ARP).
- An ARP module in the sending host takes any IP address on the same LAN as input, and returns the corresponding MAC address.
- **The ARP Query:** If the address is unknown, the sender broadcasts an ARP query packet to all other hosts and routers on the subnet to determine the MAC address.
- The node with the matching IP address replies with its MAC address, allowing the sender to finally construct the link-layer frame.

Switches vs. Routers (Section 6.4.3)

Now that we understand frames and MAC addresses, we can clearly define the difference between the two most common network devices:

- **Routers (Layer 3):** Routers are store-and-forward packet switches that forward packets using network-layer (IP) addresses. They operate the Control Plane logic.
- **Switches (Layer 2):** Although a switch is also a store-and-forward packet switch, it is fundamentally different from a router in that it forwards packets using MAC addresses.
- Switches do their job transparently—meaning the host or router happily sends the frame into the LAN without having to explicitly address the frame to the intervening switch.